**University of Hertfordshire**

**School of Engineering and Technology**

5FTC2068 Applied Robotics and AI Project

# Assignment 1:

# Brief Discussion of Add, Commit, Push, and Clone in Git

Author: Kenzo Jonathan Harapan Ji

Student ID: 4234QLMV

Lecturer: Ms Sarina Tajudin

Submission date: 20 October 2026

## 1. Add, Edit, commit, Push, and Clone

**a. Clone**. Git clone downloads the whole repository, including the full history, and remembers GitHub as the remote called origin, the command is as such:

cd $HOME\Desktop

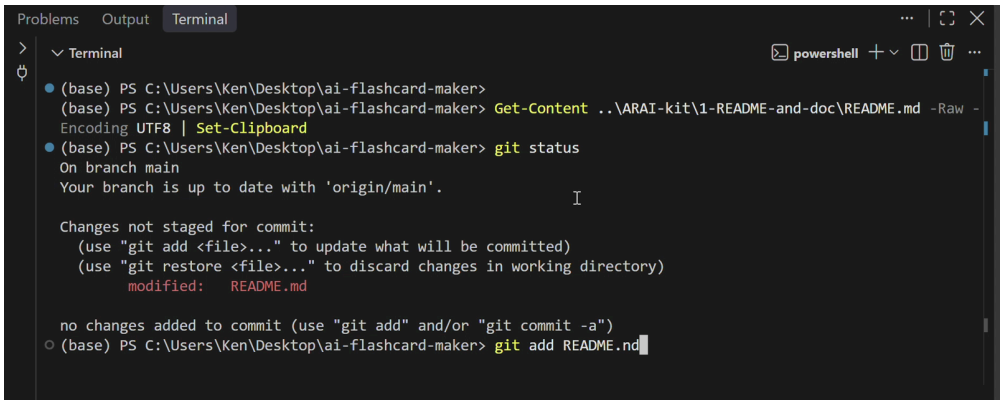
git clone https://github.com/kennnyyyyyyy/ai-flashcard-maker.git

cd ai-flashcard-maker



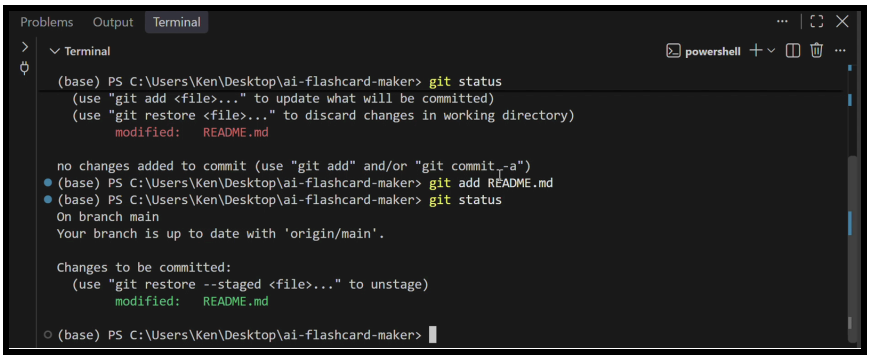

Screenshot 7: Cloning the repository

**b. Edit**. I replaced the default README with a full description of the project thats simple. After saving, git status shows README.md in red, which means Git noticed the change but its not staged yet.



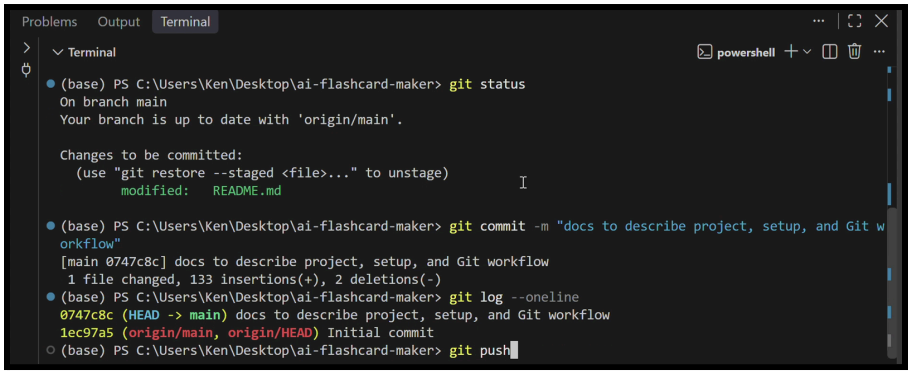

Screenshot 8: The edited README is red, not staged for commit

**c. Add**. git add README.md puts the change in the staging area. Running git status again shows its green under “changes to be committed”.



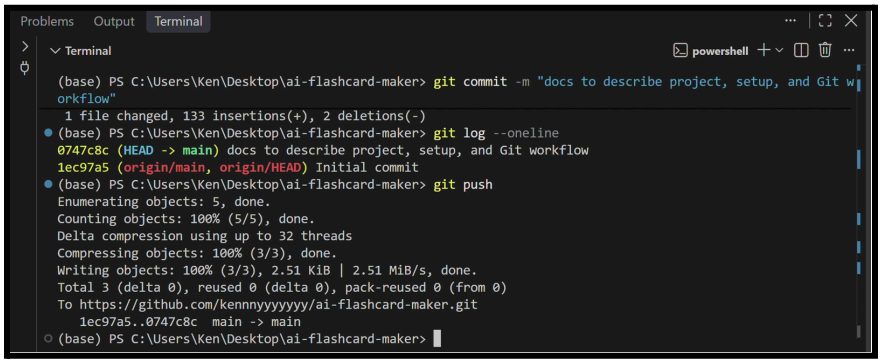

Screenshot 9: after git add, README.md is staged (green)

**d. Commit**. git commit -m "docs to describe project, setup, and Git workflow". saves the staged change as a snapshot in local history. git log --oneline shows the new commit on top of the initial commit. At this point the commit exists only locally: origin/main is still one commit behind.

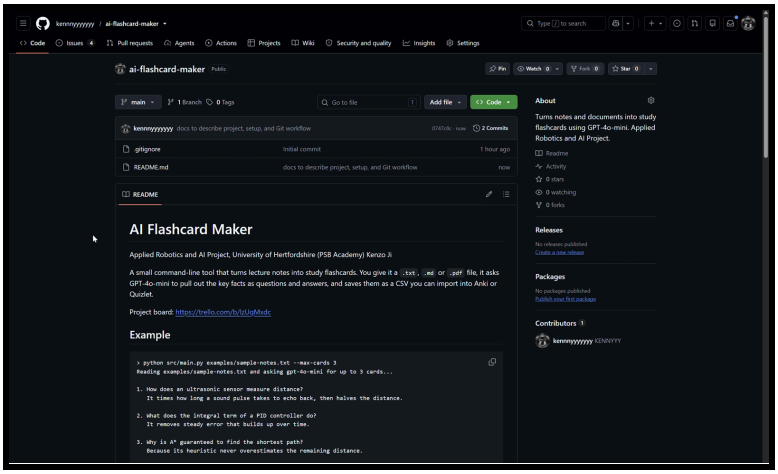



Screenshot 10: The commit and the log. As seen Head -> main is ahead of origin/main

**e. Push**. git push uploads the commit to GitHub. The output ends with “1ec97a5..0747c8c main -> main”, which means GitHub’s main now points to the new commit.




Screenshot 11: Pushing the commit to GitHub




Screenshot 12: The new README on GitHub after the push

**2. The Difference between git add, commit, push, and clone**

| | **git clone** | **git add** | **git commit** | **git push** |
|---|---|---|---|---|
| Moves work from | GitHub | Working folder | Staging area | Local repository |
| Moves work to | A new folder on my laptop | Staging area | Local repository | GitHub |
| How often I use it | Once per project | Before every commit | After each small piece of work | When the work is ready to share |
| Creates history? | No, it copies all of it | No | Yes, one new commit | No, it sends commits that already exist |
| Needs internet? | Yes | No | No | Yes |
| Can others see it? | No | No | No | Yes |

The main difference is where each command works. Clone and push only talk to GitHub, and they go in opposite directions. Clone brings the whole repository down once at the start to local, while push sends new commits up to GitHub. Add and commit only happen locally. Only commit creates history, while add just prepares the next commit, and push copies commits that already exist. Its useful to keep add and commit separated because of the ability to choose what goes into each commit.